Sequences

A SEQUENCE IS A SERIES OF NUMBERS WRITTEN AS A LIST THAT FOLLOWS A PARTICULAR PATTERN, OR "RULE."

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Each number in a sequence is called a "term." The value of any term in a sequence can be worked out by using the rule for that sequence.

The terms of a sequence

The first number in a sequence is the first term, the second number in a sequence is the second term, and so on.

▷ **A basic sequence**
For this sequence, the rule is that each term is the previous term with 2 added to it.

rule for this sequence is each term equals previous term plus 2

+2 +2 +2 +2

first term is 2

fifth term is 10

dots show sequence continues

2, 4, 6, 8, 10, ...

1st term 2nd term 3rd term 4th term 5th term

Finding the "nth" value

The value of a particular term can be found without writing out the entire sequence up until that point by writing the rule as an expression and then using this expression to work out the term.

▷ **The rule as an expression**
Knowing the expression, which is 2n in this example, helps find the value of any term.

2n

expression used to find value of term—1 is substituted for n in 1st term, 2 in 2nd term, and so on

means 2 × n

substitute 1 for n

$2n = 2 \times 1 = 2$

1st term

To find the first term, substitute 1 for n.

$2n = 2 \times 2 = 4$

2nd term

To find the second term, substitute 2 for n.

$2n = 2 \times 41 = 82$

41st term

To find the 41st term, substitute 41 for n.

$2n = 2 \times 1{,}000 = 2{,}000$

1,000th term

For the 1,000th term, substitute 1,000 for n. The term here is 2,000.

In the example below, the expression is 4n – 2. Knowing this, the rule can be shown to be: each term is equal to the previous term plus 4.

+4 +4 +4 +4

sum of 14 and 4

2, 6, 10, 14, 18, ...

1st term 2nd term 3rd term 4th term 5th term

expression here is 4 multiplied by n, minus 2

4n – 2

value of term

$4n - 2 = 4 \times 1 - 2 = 2$

1st term

To find the first term, substitute 1 for n.

$4n - 2 = 4 \times 2 - 2 = 6$

2nd term

To find the second term, substitute 2 for n.

$4n - 2 = (4 \times 1{,}000{,}000) - 2 = 3{,}999{,}998$

1,000,000th term

For the 1,000,000th term, substitute 1,000,000 for n. The term here is 3,999,998.